

ACHINTYA NIGAM

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Eligible to work in Ireland under Stamp 1G post completion of masters

Education

Munster Technological University (MTU) <i>MSc Cybersecurity</i>	Cork, Ireland <i>Sep 2026 – Expected Aug 2027</i>
Vellore Institute of Technology (VIT) <i>B.Tech Computer Science and Engineering — CGPA: 9.02/10 (First Class with Distinction)</i>	Vellore, India <i>2022 – 2026</i>

Experience

Independ <i>Junior Software Developer</i>	Vellore, India <i>Dec 2024 – Mar 2025</i>
- Engineered Kairos, a React Native client management system, implementing background processing services designed to securely handle and store sensitive customer call data. - Maintained application architecture and data integrity while optimizing automated, security-conscious data workflows for Android environments.	
Independ <i>Frontend Developer Intern</i>	Vellore, India <i>Jul 2024 – Nov 2024</i>
Developed the Next.js frontend for the Independ platform, implementing secure API data consumption and safe user-input handling.	
Alumnus Software <i>Software Engineer Trainee Intern</i>	Remote <i>Jan 2024 – Jun 2024</i>
- Developed scalable Spring Boot and React.js applications with a strong focus on secure backend architecture. - Secured REST API endpoints and user sessions by implementing JWT-based authentication and bcrypt password hashing.	

Cybersecurity Projects & Research

TCP/IP Network Traffic Analysis and Packet Crafting	Linux, tcpdump, Wireshark, hping3, Netcat
Crafted and analyzed TCP/IP packets using Wireshark, tcpdump, and hping3, investigating differences in Windows ICMP behaviors and undocumented hping3 APD packet construction behaviors. Technical Write Up: Medium Article	
Post-Quantum Cryptography on Dual-Core Microcontrollers	C, RP2350A, Raspberry Pi Pico 2 W, Kyber, Dilithium
- Optimised CRYSTALS-Kyber and CRYSTALS-Dilithium cryptographic operations using the dual-core RP2350A architecture, achieving approximately 30% reduction in key-generation time for Kyber. Quantified Performance vs Energy Trade-Offs. - Co-authored the resulting research as First Author, published in IEEE Access DOI: 10.1109/ACCESS.2026.3677890	
SOC Analyst Pathway	Hack The Box
- Used Splunk Processing Language (SPL) to query event logs, hunt for statistical outliers, and investigate potential network intrusions within enterprise environments. - Analyzed Windows Sysmon event IDs and conducted foundational reverse engineering of x86 assembly routines to identify and trace malware behavior.	

Technical Skills

Programming: Python, C, C++, Java, JavaScript, TypeScript
Cybersecurity: Network Security, TCP/IP, Vulnerability Assessment, Malware Analysis, SIEM, Security Monitoring
Security Tools: Wireshark, tcpdump, hping3, Nmap, Splunk, Sysmon
Web / Backend: React, Next.js, Spring Boot, REST APIs, HTML/CSS
Cloud / Databases: AWS, MySQL
Systems / Tools: Linux, Git

Certifications

AWS Certified Solutions Architect – Associate	2026
AWS Certified Cloud Practitioner	2024